Move Over Silicon

The fundamental building blocks of circuits—transistors themselves—are in for a maekover

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"The chip industry is facing a roadblock"—that's something we've been hearing for some time now in various forms. What we've also been seeing for some time now is that these "roadblocks" never seem to do anything—someone finds a way to get round them, and then there's another roadblock. Nothing exciting seems to be happening, as in "machines will *not* get any faster after 2010!" The tech juggernaut rolls on.

However, there is one thing common to most of the problems hardware researchers face: it's always about miniaturisation. Things are always getting smaller and more powerful (as if you didn't know that)—but the remarkable thing is that Moore's law or Kryder's law (a sort of Moore's law for hard disks) or whatever law keep on holding true.

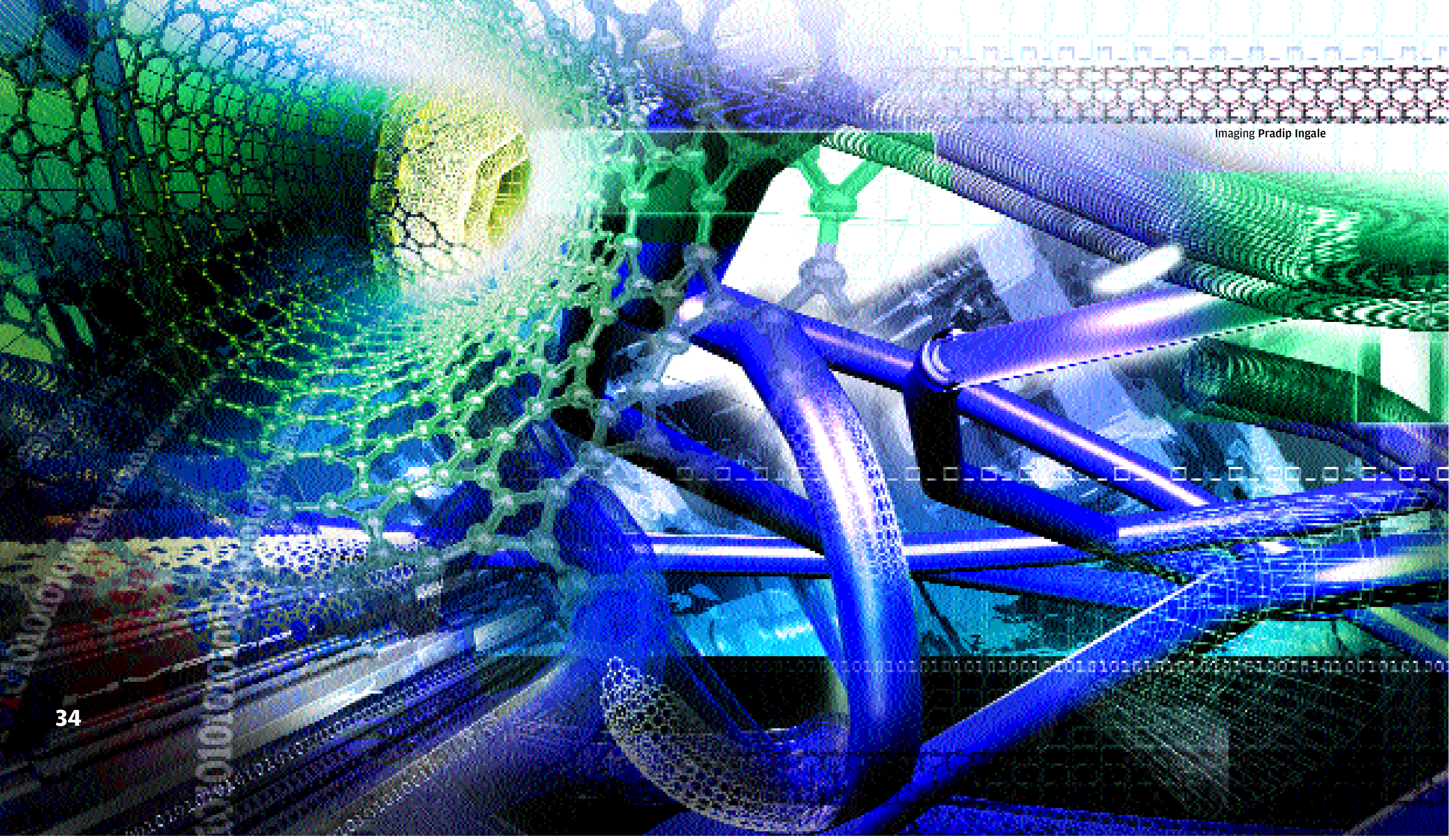
Tech 2020

We can't resist talking about Moore's "law" here. We say "law" in quotes because it isn't, of course, a law per se: it's an observation, one that says that the number of transistors on a chip will double every eighteen months. Or should that be "two years"? And "the density of components on a chip"? It's not important—it's only an academic exercise in history to find out what Gordon Moore originally said. What's important is that things will shrink exponentially, and that usually means that capacities and speeds and such will increase exponentially as well.

Now there's the 2018 thing to contend with. Industry analysts predict that Moore's law will hold good only until 2018, or generally, around the year 2020. Their point: that the shrinkage of silicon transistors cannot continue indefinitely, and that by around 2020, we'd have reached the stage where they cannot be shrunk anymore. But that's about *silicon*—what about other materials? Enter silicon's cousin in the periodic table, *carbon*.

The Stuff Soot Is Made Of

NEC's Sumio Iijima is reported to have been the first to observe, in 1991, carbon nanotubes. He saw them in electron microscope images of—of all things—soot. The nanotubes were essentially sheets of graphite rolled into concentric cylinders—one cylinder inside another; such structures are called multi-walled nanotubes. And in 1993, Iijima and IBM's Don Bethune found, in research independent of each other, that they could produce *single-walled* nanotubes, that is, no tubes within each other. These consist of a single atomic layer of the graphite structure, and look something like a hexagonal wire-



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